

Disassembly task planning for end-of-life automotive traction batteries based on ontology and partial destructive rules

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ABSTRACT

Disassembly is an essential step for the cascade utilization of end-of-life automobile power batteries. The diversity in types and structures of end-of-life automobile power batteries has led to much waste in time and cost in manual disassembly processes. With the incoming large-scale retirement of automobile power batteries, it is urgent to use artificial intelligence technology to enable the automation of battery disassembly planning. In order to establish a complete and open product information model to realize the automatic disassembly task planning of end-of-life automobile power battery, a disassembly task planning method of automobile power batteries is proposed based on ontology and partial destructive rules. This presented approach is in combination with case-based reasoning/rule-based reasoning, which is utilized as the mechanism for disassembly knowledge reuse and reasoning. Firstly, a disassembly information ontology of automobile power batteries is constructed to describe the components information and assembly relation. Then, a set of partial destructive rules are formulated to guide the dismount of parts with destructive connection. Thirdly, a disassembly sequence generation method is presented to infer feasible planning schemes from the rule base. Finally, the effectiveness of the method is tested with a case study of a power lithium-ion battery pack. The case study has indicated that this presented method can generate the disassembly task schemes quickly and effectively, when applied to the disassembly of large-scale heterogeneous automobile power batteries.

1. Introduction

In response to increasingly severe global warming, countries are actively laying out their plans to achieve the goal of "carbon neutrality". The transition to vehicle electrification has become an important initiative in the world [1]. China is the world's largest electric vehicle producer and market in the world, with 1.367 million new energy vehicles sold in 2020, accounting for 42.19 % of the world's total [2]. By the end of 2020, the number of pure electric vehicles in China had reached 4 million, accounting for 81.32 % of the total number of new energy vehicles [2]. With such many electric vehicles in service, numerous batteries will be retired a few years later and the industry will face massive challenges in reuse of used batteries [3]. Battery pack is an expensive component of electric vehicles. It consists of battery modules,

Battery management system (BMS), cooling system, power electronic devices and harmful chemicals [4]. Reuse of these batteries not only retain the value of reusable components, but also keep the valuable rare materials. Therefore, the processes to enable the maximum value retention of automotive traction batteries (ATBs) reduces environmental pollution and the demands of new resources [5,6].

The proposed method to enable maximum value retention of ATBs firstly considers product recovery via cascade utilization and then material recovery. The so-called cascade utilization refers to the application scenario where the battery is disassembled into modules or cells which are re-evaluated and restructured, and then the secondary use is determined according to the attenuation degree of battery capacity [7]. Disassembly is instrumental for product recovery [8]. In order to retain the value of used ATBs efficiently, disassembly is a necessary and

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significant step [9]. However, it faces many challenges (1) Currently, there is no design standard for battery packs/modules, resulting in a variety of battery types, sizes, assembly structures and battery chemical compositions, as a result, the disassembly process is ad-hoc; (2) Defects such as deformation and corrosion of the used battery packs affect the implementation of disassembly task. The connection between the battery cells is generally welded. If the positioning of the cutting cell is not accurate, the tool may penetrate the cell, resulting in electrolyte overflow or explosion [10]. Due to the variety and varying defects on batteries. If machine vision technology could be used to identify the batteries of different specifications and the actual state of their components, and the previous disassembly knowledge can be applied to the disassembly system, then it is possible that the optimal disassembly task scheme (DTS) can be automatically generated to solve the challenge (1). In combination with advanced visual recognition technology and control technology to adjust the cutting position and realize precise cutting operation, the optimal DTS can be applied to solve the challenge (2). Therefore, via integrating artificial intelligence technology with the disassembly system to improve the automation of battery disassembly, the proposed method aims to overcome the challenges.

Although it is very difficult to realize the intelligent disassembly of ATB at both technical and economic levels at present, it has a good prospect in solving the problem of ATBs disassembly [1,11] in near future when the number of end-of-life ATBs is large, then the intelligent disassembly can not only improve the disassembly efficiency and safety, but also reduce costs. An important research content of intelligent disassembly is intelligent disassembly task planning (DTP). It refers to the rapid and flexible generation of the optimal disassembly scheme including the sequence, direction and tool to be used for the parts disassembly based on the information identified by machine vision technology and previous disassembly knowledge, which is essential for the rapid implementation of the subsequent disassembly process [10]. As we know, the components in the battery pack that have the highest value are modules and cells. In addition to non-destructive connections in battery packs, potentially destructive connections exist. For instance, cells are often welded to a conductive busbar. Modules may be fastened by bolts, and the cables connected with the modules are fixed by ties. Modules and the bottom shell are generally fixed by glue. For such batteries containing parts with destructive connections (the parts that are unable to be disassembled theoretically (UTD) namely, UTD parts) [12], a partial destructive way is needed to remove these UTD parts. The method of DTP based on ontology and partial destructive rules is proposed in this paper, which reuses and shares previous disassembly knowledge to generate feasible DTSs automatically and flexibly.

The organizational structure of this paper is as follows: Section 2 summarizes the relevant research work. Section 3 shows the overall plan of DTP. Then the disassembly ontology and product semantic model are constructed in Section 4. Section 5 describes the hybrid method of DTP based on case-based reasoning/rule-based reasoning (CBR/RBR). Case studies and discussion in Section 6. Section 7 summarizes and prospects the full text.

2. Literature review

Since the 1990s, with the enhancement of human environmental awareness, disassembly sequence planning (DSP) has attracted extensive concern [13–15]. This section discusses the state of the art of disassembly of ATBs, product information model, knowledge reuse and

reasoning mechanism.

2.1. Disassembly of ATBs

Since 2018, ATBs have been gradually retired on a large scale [16]. In recent years, scholars have begun to study the disassembly of ATBs from the following aspects: reuse decision-making and evaluation [4, 17–19], tools for auxiliary disassembly [20–22], intelligent disassembly [10,11,23–26] and DSP. The research focus of intelligent disassembly is the design and implementation of human-machine cooperative disassembly workstation and the design of intelligent disassembly system with advanced technologies. However, there are few studies on DSP of ATBs. For example, Wegener et al. [27] proposed a priority matrix and manually derived a feasible disassembly sequence. Ke et al. [28] established a framework-subgroup structure and generated the optimized disassembly sequences using a hybrid of disassembly matrix and an improved genetic algorithm. However, there is little research on automatic disassembly task planning of ATBs. With the sharp increase in end-of-life of ATBs, more and more researchers realize that it is imperative to implement intelligent disassembly that is enabled by the automated disassembly task planning of ATBs.

2.2. Product information model

Product information model represents product and disassembly information in a structured form, which is the basis for the study of DSP [29]. In order to build an automatic disassembly system, a complete and open product information model [30] is needed to provide a shared, extensible and organized information structure in the specified format. At present, there are two main types of models widely used, accounting for more than 70 % of the published papers on DSP [31]. One is the graph-based model, which can intuitively show all feasible disassembly sequences. Ren et al. [32] used a disassembly precedence graph to model and expressed the priority relation between parts with a precedence matrix. Tian et al. [33] used AND/OR graph to model and described the priority and/or relations between parts with four matrices. The other is the matrix-based model. For example, Smith and Chen [34] used four matrices to express the geometric constraints and disassembly directions between adjacent parts only without the need to represent the geometric constraints between all parts, which greatly reduced the computational time and space. Rickli and Camelio [35] used a priority matrix, a cost matrix and an environmental impact matrix to describe the feasibility, cost and environmental impact of disassembly operations, respectively. Tao et al. [36] established a dynamic disassembly precedence matrix composed of four sub matrices to describe the disassembly direction and the relations between components, but the disassembly direction required to be encoded. Although the above two types of models can well describe the disassembly priority relation and connection relation between parts. These models can be constructed by computers in matrix or other numerical forms, however, there is a lack of data semantics such as concepts and relations in the disassembly field in these models. This makes it difficult to reuse past successful disassembly schemes and expand this existing disassembly knowledge and to represent other semantic information such as disassembly tools. Moreover, when there is a lack of information about disassembly direction and disassembly tools in a disassembly plan, the possibility of automatic execution of the disassembly scheme will be limited.

Based on this, researchers customized some product information

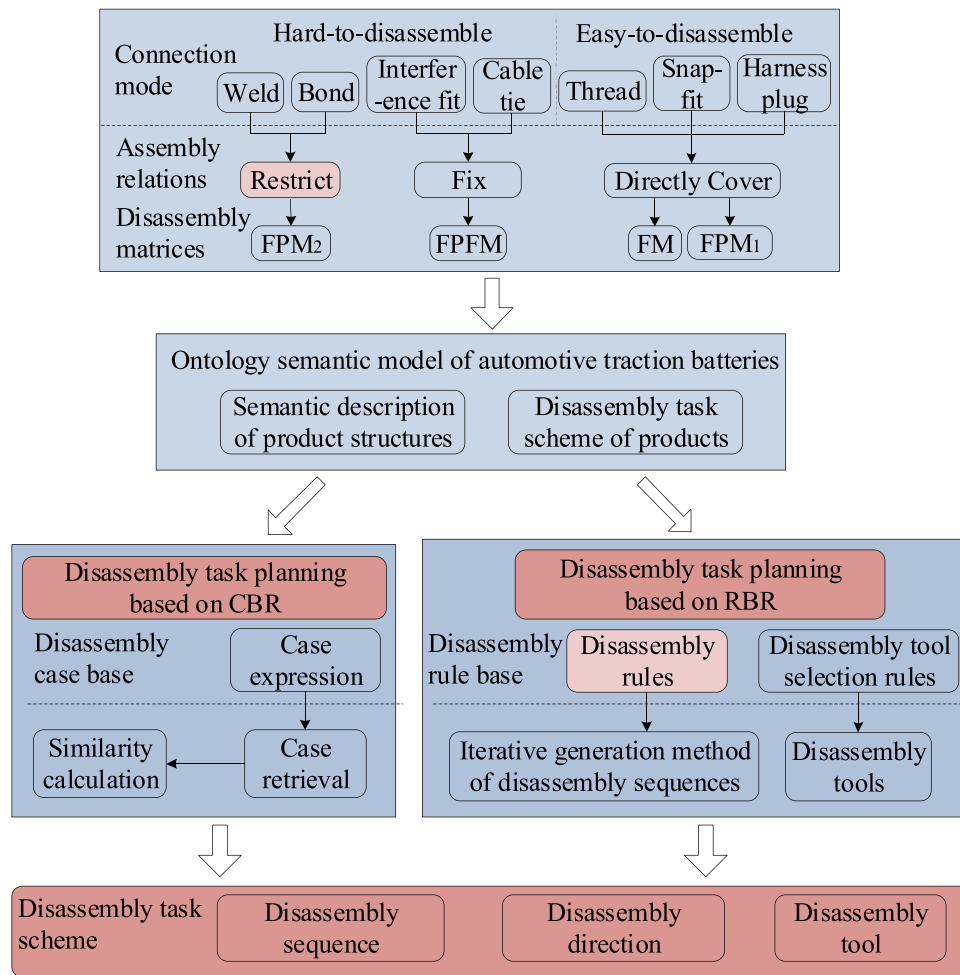


Fig. 1. Framework of the disassembly task planning method for ATBs.

models to contain additional information, for example, Disassembly information model (DIM) in which product, process, fuzzy semantics and retirement knowledge are represented [37,38]. The Disassembly semantic model (DSM) in existing research can show knowledge of product hierarchy, assembly relations, disassembly cases, disassembly rules and disassembly costs [39], but there is a lack of information such as disassembly tools in these DSM models. Product ontology model (POM) can represent knowledge such as product hierarchy, connection and constraint relations, disassembly rules and disassembly tool selection rules [40], but the existing disassembly scheme cannot be reused. The above models have built sharable, extensible information structures using ontology in disassembly domain. However, DIM is a general disassembly model with complex structure and many aspects need to be improved. The existing DSM and POM are often the disassembly models for electromechanical products, and their disassembly rules are only applicable to non-destructive disassembly products, but not to ATBs.

2.3. Knowledge reuse and reasoning mechanism

Ontology can be used to build conceptual models and express semantic knowledge, which has been widely used in artificial intelligence, system engineering and other fields [41,42]. It can unify and manage disassembly information at different lifecycle stages and build an organized and interconnected disassembly knowledge system [43]. Jiang et al. [40] used ontology and ontology rules to represent and expand disassembly knowledge, but reuse of existing disassembly cases

including skills and experience are not considered, and the value of disassembly knowledge is not recognized. In order to enable the rapid generation of disassembly tasks for products to be disassembled, CBR is used to retrieve and reuse the stored disassembly cases. The following studies reveal that the combination of ontology and CBR achieves a good reuse of case knowledge. Park et al. [44] proposed a reasoning system of ship collision avoidance using fuzzy reasoning to represent the cases stored in ontology. Bouhana et al. [45] designed an information retrieval system based on ontology and CBR to bring about the itinerary search of urban freight users. Camarillo et al. [46] combined ontology, CBR and product lifecycle management to propose a multi-agent system to solve manufacturing problems. RBR could be used to improve the reusability of knowledge in the absence of matching cases. Yang and Hsu [47] designed a three-tier agent mechanism based on ontology and CBR/RBR to query and predict solutions. Chen et al. [39] put forward a method combining ontology with CBR/RBR to implement the planning of disassembly sequence and disassembly direction for electromechanical products. Xu et al. [48] presented a fault diagnosis method for loaders based on ontology and CBR/RBR. The above research reveals the effectiveness of combining ontology with CBR/RBR, which can realize the integration, reuse and reasoning of knowledge.

In conclusion, in order to solve the disassembly problem of large-scale and structural diversification of ATBs, reuse of the existing disassembly knowledge in disassembly planning process can improve the flexibility of disassembly scheme and reduce disassembly cost and time. To reuse the existing disassembly knowledge and realize the intelligent

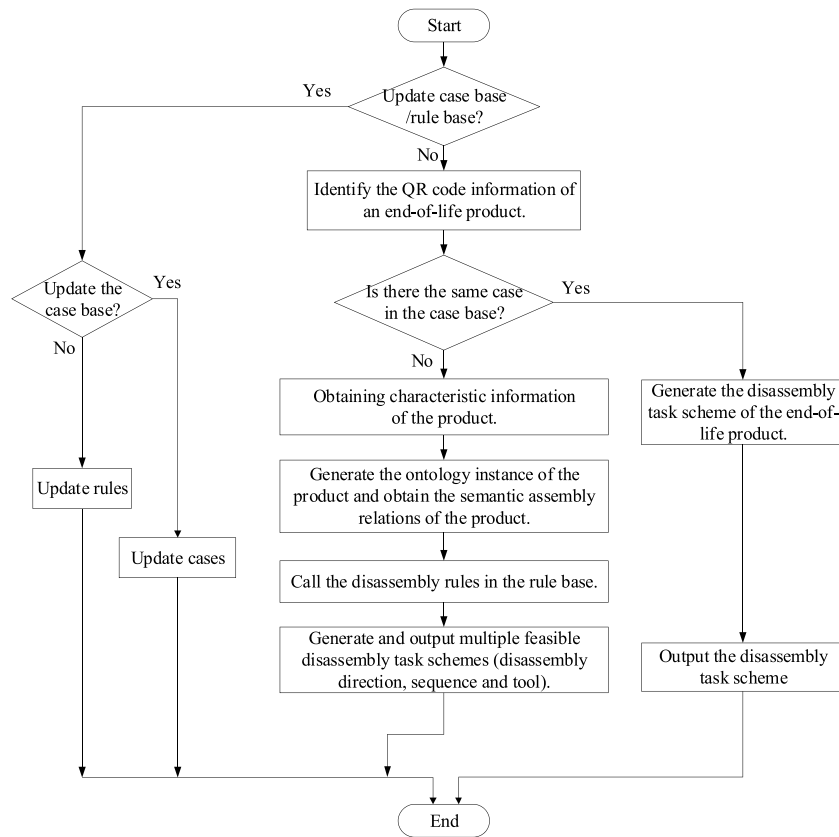


Fig. 2. Flow chart of disassembly task planning for ATBs.

disassembly of ATBs, it is necessary to build a shareable and extensible product information model. But nowadays, there are few studies on the disassembly ontology model and UTD issue of ATBs. Therefore, on the basis of DSM and POM models, this paper establishes a disassembly ontology model for ATBs, and proposes a method of DTP based on ontology and partial destructive rules. The contributions of this method to knowledge are as follows:

- (1) An ontology model supporting case/rule reasoning is constructed, which is dedicated for the disassembly of battery packs and modules.
- (2) A set of partial destructive disassembly rules and disassembly tool selection rules are designed to solve the UTD problem.
- (3) Considering the normal direction of part disassembly, an iterative generation method of parallel disassembly sequence is designed to eliminate unrealistic disassembly sequences. The disassembly task includes disassembly sequence, disassembly directions and disassembly tools.

3. The overall method

In order to facilitate machine vision to identify vehicle power battery information, China has introduced relevant policies. For example, since August 2018, in China, a two-dimensional code (QR code) containing the design and production information must be affixed to all packs/modules/cells of ATB. Enterprises need to upload the whole process

information of battery production, sales, use, end-of-life, recycling to the specified platform [49]. The policy will benefit greatly the battery recycling. A DTP method of ATBs based on ontology and partial destructive rules is proposed in combination with QR code technology. The framework of this method is shown in Fig. 1. First, a variety of assembly relations are summarized and described in the form of matrices. Then, according to these disassembly matrices, the ontology semantic model is constructed. Moreover, the disassembly case base is established to implement the DTP based on CBR. When there is no matching case, the disassembly rule base is built to perform the DTP based on RBR.

The flow chart of DTP for ATBs is shown in Fig. 2. Firstly, it determines whether to update the case library/rule library. If not, scan the QR code on the battery to obtain design and production information and implement the DTP based on CBR. If the case matching is successful, the DTS is output directly, otherwise, the DTP based on RBR is performed. As the information stored in QR codes is limited, it needs to collect battery structure and assembly information manually or employing machine vision technologies to generate ontology instances and semantic assembly relations. Then, feasible disassembly sequences and disassembly tools may be obtained by adapting relevant disassembly rules that are stored in the rule base. It should be noted that the library of disassembly rules can be supplemented and updated.

4. Disassembly ontology and product semantic model

In this section, a set of disassembly matrices is used to represent the

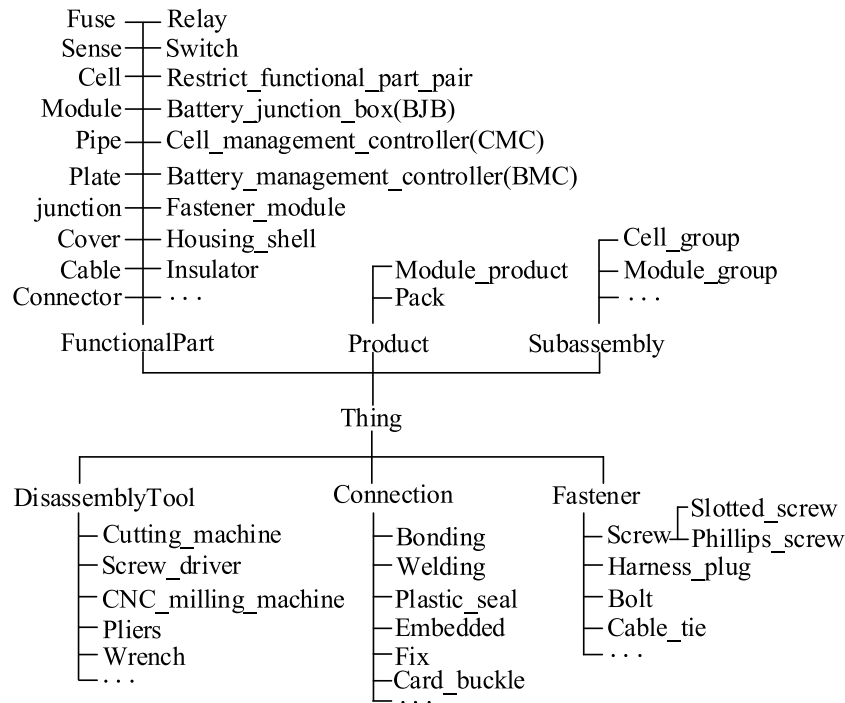


Fig. 3. Classes in the disassembly ontology of ATBs.

assembly relation of ATBs, which can be used to describe the object attributes and data attributes of ontology, so as to build the disassembly ontology model. On this basis, the semantic model of ATBs is designed.

4.1. Disassembly matrix

4.1.1. Assembly relations

The minimum unit of ATB products is divided into fixing parts and functional parts to facilitate the construction of disassembly ontology. Pressure parts and restricted functional part pair are also defined as required, as follows.

Functional part: The part plays a certain role and has effect on the function of ATBs, which can be understood as any other parts except fixing parts in a broad sense.

Fixing part: The part plays the role of connecting or fixing functional parts/subassembly, such as card buckle, cable tie, pressure part, and threaded fastener. For instance, when the wiring harness and cable are connected with functional parts, the wiring harness and cable are considered as fixing parts.

Pressure part: The non-shaft part with interference fit assembly relation with a functional part is regarded as a kind of fixing parts that needs to be dismantled first. If the pressure part is formed by welding, then the interference fit relation is removed by removing the welding relation. For instance, the metal shell of modules in case study.

Restricted functional part pair: It refers to two functional parts connected by welding or bonding, and there is a restrictive relation between them.

This paper abstracts three assembly relations: *Fix*, *Directly cover* and

Restrict, which are described as follows:

Fix: *fix* shows the assembly relation between fixing parts and functional parts. "A functional part is fastened in a certain direction by another fixing part," then their disassembly order should be "fixing part > functional part" [50], such as thread connection, card buckle connection, plug connection, tie connection, interference fit.

Directly cover: *directly cover* expresses the assembly relation between fixing parts/functional parts and functional parts. "A fixing part 1 or a functional part 1 is directly covered by functional part 2 in a certain direction" means that there are no other functional parts between the fixing part 1 or the functional part 1 and functional part 2 in the orientation. Then, their disassembly order should be "functional part2-> fixing part 1/functional part 1" [50].

Restrict: *restrict* represents the assembly relation between functional parts. "Functional part 1 and functional part 2 restrict connection in a certain direction," then their disassembly order should be "functional part 1 < -> functional part 2". That is, the two functional parts are disassembled at the same time without sequence, such as two functional parts that are joined by welding and bonding.

The above method is also suitable for expressing the assembly relations between subassemblies. At this point, just consider the subassembly as a functional part.

4.1.2. Disassembly matrix

A set of disassembly matrices is proposed based on [39] to represent the three assembly relations. They succinctly express the assembly relations between fixing parts and functional parts.

(1) FM

Table 1
Object properties in the disassembly-ontology.

No.	Object Property	Doman	Range	Inverse Property
1	directCover_plusX	FunctionalPart, Subassembly	Fastener, FunctionalPart, Subassembly	isDirectCoveredBy_minusX
2	isDirectCoveredBy_plusX	Fastener, FunctionalPart, Subassembly	FunctionalPart, Subassembly	directCover_minusX
3	directCover_plusY	FunctionalPart, Subassembly	Fastener, FunctionalPart, Subassembly	isDirectCoveredBy_minusY
4	isDirectCoveredBy_plusY	Fastener, FunctionalPart, Subassembly	FunctionalPart, Subassembly	directCover_minusY
5	directCover_plusZ	FunctionalPart, Subassembly	Fastener, FunctionalPart, Subassembly	isDirectCoveredBy_minusZ
6	isDirectCoveredBy_plusZ	Fastener, FunctionalPart, Subassembly	FunctionalPart, Subassembly	directCover_minusZ
7	fix_plusX	Fastener	FunctionalPart, Subassembly	isFixedBy_minusX
8	isFixedBy_plusX	FunctionalPart, Subassembly	Fastener	fix_minusX
9	fix_plusY	Fastener	FunctionalPart, Subassembly	isFixedBy_minusY
10	isFixedBy_plusY	FunctionalPart, Subassembly	Fastener	fix_minusY
11	fix_plusZ	Fastener	FunctionalPart, Subassembly	isFixedBy_minusZ
12	isFixedBy_plusZ	FunctionalPart, Subassembly	Fastener	fix_minusZ
13	restrict_X	FunctionalPart	FunctionalPart	restrict_X
14	restrict_Y	FunctionalPart	FunctionalPart	restrict_Y
15	restrict_Z	FunctionalPart	FunctionalPart	restrict_Z
16	is_part_of	Fastener, FunctionalPart, Subassembly	Subassembly, Product	has_part_of
17	has_disassembled_after	Fastener, FunctionalPart, Subassembly	Fastener, FunctionalPart, Subassembly	N/A
18	has_connection	Fastener, FunctionalPart, Subassembly	Connection	N/A
19	is_removed_with	Fastener, FunctionalPart, Subassembly	DisassemblyTool	N/A

FM defines the directly cover assembly relation between fixing parts and functional parts. It records the numerical codes of the direct covering functional parts for each fixing part in the detachable orientation. The direct covering functional parts indicate that there are no other functional parts in the space between a fixing part and its direct covering functional parts on the disassembly direction of a fixing part. In the matrix, columns represent six disassembly directions: +X/-X/+Y/-Y/+Z/-Z, and rows represent each fixing part. N denotes the functional part number.

$$FM = \begin{cases} N, & \text{thenumerical codes of thedirect covering} \\ & \text{functional parts in the detachable direction.} \end{cases}$$

functional parts in the detachable direction. 0, no direct covering functional

part in the detachable direction. part in the detachable direction. $\emptyset$, others

(2) FPFM

FPFM defines the fixing assembly relation between functional parts and fixing parts. It records the numerical codes of the fixing parts which are used to fasten functional parts in each disassembly direction. In the matrix, columns represent disassembly directions, and rows represent each functional part.

$$FPFM = \begin{cases} N, & \text{the numerical} \\ & \text{codes of the fixing parts which are used to fix functional parts.} \end{cases}$$

$\emptyset$, others.

(3) FPM₁

FPM₁ defines the direct cover assembly relation between functional parts. It records the numerical codes of the direct covering functional parts for each functional part in each disassembly orientation. In the matrix, columns represent disassembly directions, and rows represent each functional part.

Table 2
Data properties in the disassembly-ontology.

No.	Data Property	Domain	Range	Attribute Description
1	disassembledDirection_plusX disassembledDirection_minusX disassembledDirection_plusY disassembledDirection_minusY disassembledDirection_plusZ disassembledDirection_minusZ	Fastener, FunctionalPart, Subassembly	boolean	Record the disassembly direction of a component.
2	normalDirection_X normalDirection_plusY normalDirection_Z	FunctionalPart, Subassembly	boolean	Record the normal direction of a component (regardless of the-Y direction).
3	type_code	Product	string	Record product type code: battery pack-P, module-M, cell- C.
4	material_code	Product	string	Product material type code: A~G.
5	vendor_code	Product	string	Product manufacturer code.
6	model_code	Product	string	Product specification model code.
7	trace_information_code	Product	string	Product traceability information code.
8	date_code	Product	string	Product production date code.
9	serial_number	Product	unsigned long	Product serial number.
10	cascade_utilization_code	Product	string	Product cascade utilization code (RP/RM/RC).
11	id	Fastener, FunctionalPart, Subassembly, Product	string	Record the identity of a product /component.
12	component_name	Fastener, FunctionalPart, Subassembly	string	Record the name of a component.
13	quantity	Fastener, FunctionalPart, Subassembly	int	Record the number of components of the same type.

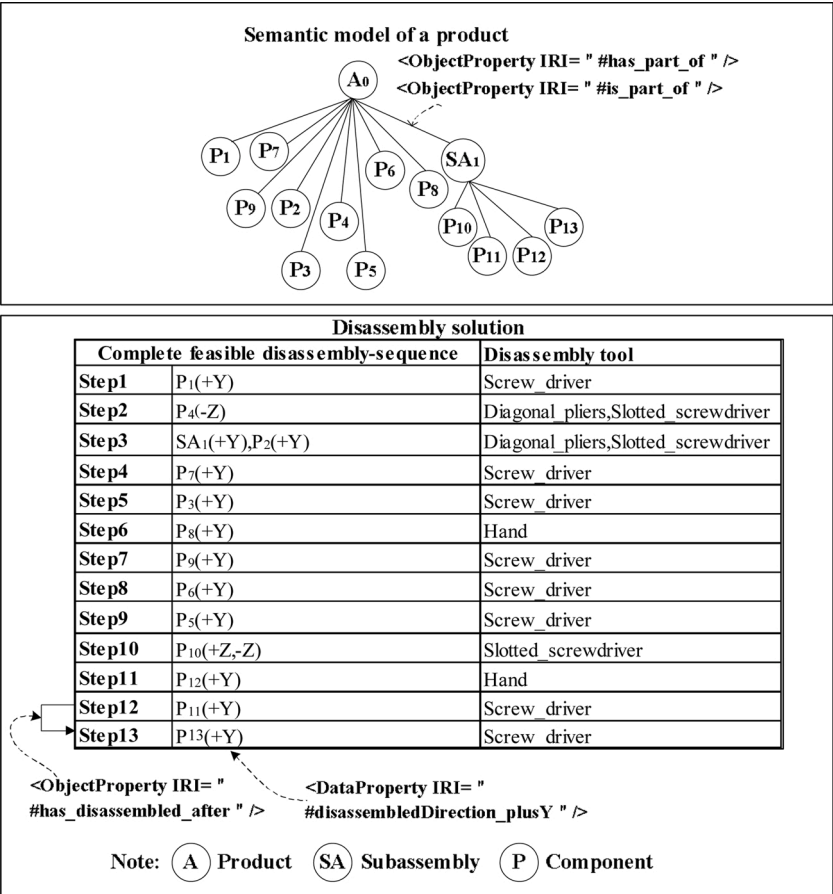


Fig. 4. Schematic of a disassembly case.

Table 4 (continued)

No.	SWRL rules	Description of disassembly rules
23	isDirectCoveredBy_plusZ(?fp,?fp1)->has_disassembled_after(?fp,?fp1) FunctionalPart(?fp), FunctionalPart(?fp1), disassembledDirection_minusZ(?fp,1), isDirectCoveredBy_minusZ(?fp,?fp1)->has_disassembled_after(?fp,?fp1)	
24	FunctionalPart(?fp), FunctionalPart(?fp1), restrict_X(?fp,?fp1)-> has_disassembled_after(?fp,?fp1), has_disassembled_after(?fp1,?fp) FunctionalPart(?fp), FunctionalPart(?fp1), restrict_plusY(?fp,?fp1)-> has_disassembled_after(?fp,?fp1), has_disassembled_after(?fp1,?fp) FunctionalPart(?fp), FunctionalPart(?fp1), restrict_Z(?fp,?fp1) ->has_disassembled_after(?fp,?fp1), has_disassembled_after(?fp1,?fp)	If the functional part A and the functional part B are restricted each other in a certain direction (except-Y direction), then the functional part A and the functional part B can be disassembled simultaneously.
25		
26		

Table 5

Symbols used in this section.

Symbol	Meaning
SetA	The collection that contains all the fixed parts and functional parts of the product.
SetB	The collection that contains all the functional parts of the product.
R _i	Represents the restricted functional part pair with subscript i, i=1,2,...n.
Set R	The collection that contains all the restricted functional part pair.
N	Number of elements of Set R.
RFPk	Represents the functional part with subscript k that make up a restricted functional part pair, k=1,2.
SA _j	Represents the subassembly with subscript j, j=1,2,...m.
Set SA	The collection that contains all the subassemblies.
M	Number of elements of Set SA.
F	Represents the fixed parts.
FP	Represents the functional parts.
PRFP	Priority disassembly of functional part.
PRF	Priority disassembly of fixed part.
Φ	Represents an empty set.

contains design information and production information. Design information includes manufacturer code, product type code, battery material type code, model code and traceability information code. Production information includes production date code, serial number and cascade utilization code. Therefore, the definition of data attributes that contains both design information and production information stored in the QR code is shown in Table 2.

4.2.2. Product semantic model

The ontology semantic model of ATBs is divided into two sections: (1) product framework

(including parts information, hierarchical structure, assembly structure and connection relations), (2) the DTS of the product (parallel disassembly sequence, disassembly direction and disassembly tool). The semantic model of the battery to be disassembled contains only the first section, while the semantic model of the disassembly case contains both. The ontology of a typical disassembly case is shown in Fig. 4.

For the first part, the class and object properties (*is_part_of*, *has_part_of*) are used to construct the hierarchy structure of products. The product structure also contains detailed information of each subassembly/component in the product (See Tables 1 and 2).

For the second part, the complete and feasible disassembly sequence of a product can be expressed by object property *has_disassembled_after* and data properties (*disassembledDirection_plusX/minusX/plusY/minusY/*

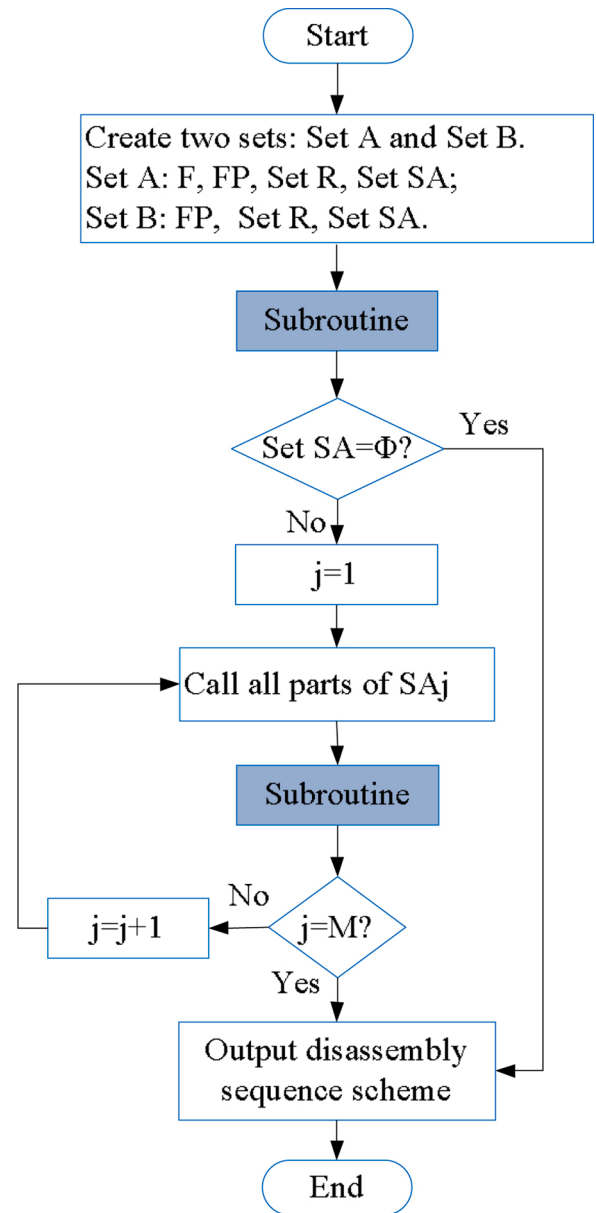


Fig. 5. Flowchart of the main program for the generation of feasible disassembly sequences.

plusZ/minusZ, *normalDirection_X/plusY/Z*). Besides, the disassembly tools and equipment used to execute the sequence can be expressed by using the object properties *is_removed_with* and *has_connection*.

5. DTP based on CBR/RBR

The CBR/RBR method is used as the reuse and reasoning mechanism of disassembly knowledge in this paper. Firstly, matched cases of previously reused are retrieved for end-of-life products. Due to the confidentiality of the battery manufacturer's information on the assembly structure of ATBs, it is hard to modify the scheme of similar cases. Therefore, if the match is successful, the DTS that matches the case can be used directly. Otherwise, a new DTS is obtained according to the partial destructive disassembly rules and disassembly tool selection rules.

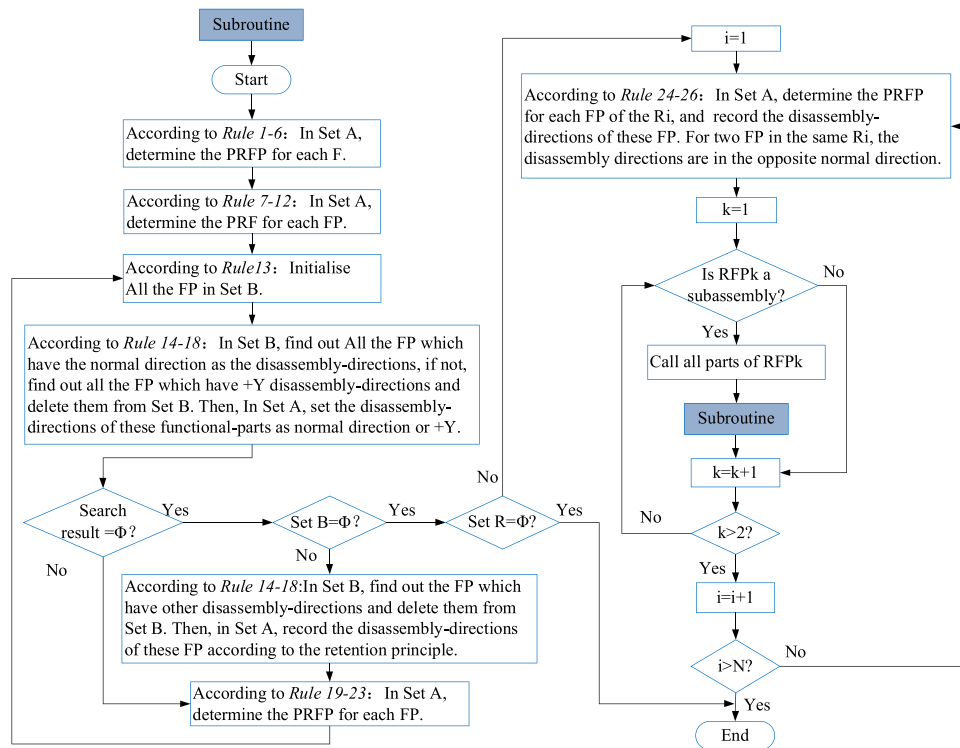


Fig. 6. Flowchart of the subroutine program for the generation of feasible disassembly sequences.

5.1. DTP based on CBR

Usually a case has many data attributes. The data attributes that have an important influence on case retrieval are selected as the influencing factors to reduce the complexity of case retrieval. CBR of ATBs is determined by five influencing factors, namely, type_code (C_1), material_code (C_2), vendor_code (C_3), model_code (C_4) and secondary_use_code (C_5). These influencing factors are described by numbers and letters. They belong to string types. Then the formula for calculating the local similarity between the influencing factors is:

$$\text{Sim}(C_i^T, C_i^S) = \begin{cases} 1, & C_i^T = C_i^S \\ 0, & C_i^T \neq C_i^S \end{cases} \quad (1)$$

In the formula, if the character/string described by the influence factor C_i of the target case T and the source case S in the case library is exactly the same, the local similarity is assigned to 1. Otherwise it is 0.

According to the local similarity of the influencing factors of disassembly cases, the overall similarity between T and S can be calculated by Eq. (2):

$$\text{Sim}(T, S) = \sum_{i=1}^m \omega(C_i) \text{Sim}(C_i^T, C_i^S) \quad (2)$$

Where $\omega(C_i)$ is the weight coefficient of the influencing factor C_i . $\omega(C_i) \in [0, 1]$, and $\sum_{i=1}^m \omega(C_i) = 1$.

AHP (Analytic Hierarchy Process) is used to construct the judgment matrix by comparing the importance of different influencing factors on

Table 6

Connection type and disassembly tool of ATBs.

No.	Connection type	Disassembly tool	Tool number
1	Bolt	Socket wrench/Double offset ring spanner	T1/T2
2	Slotted screw	Slotted screwdriver	T3
3	Phillips screw	Phillips screwdriver	T4
4	Card buckle	Slotted screwdriver	T3
5	Harness plug	Needle-nose pliers /Slotted screwdriver	T5/T3
6	Cable tie	Diagonal pliers	T6
7	Bonding	Slotted screwdriver	T3
8	Welding	CNC milling machine/ Cutting machine	T7/T8
9	Plastic seal/embedded	Cutting machine	T8

the disassembly decision of ATBs. Table 3 is the judgment matrix for the case study. Moreover, the weight coefficient of each influencing factor is determined based on Table 3, and the consistency ratio $CR = 0.0015691 < 0.1$.

If the overall similarity of between T and S is $\text{Sim}(T, S) > \varepsilon$ (similarity threshold), the task scheme of S can be used as the task scheme of T. If the overall similarity $\text{Sim}(T, S) \leq \varepsilon$, the case is not adjusted and the task scheme is obtained by RBR. This paper sets $\varepsilon = 0.95$.

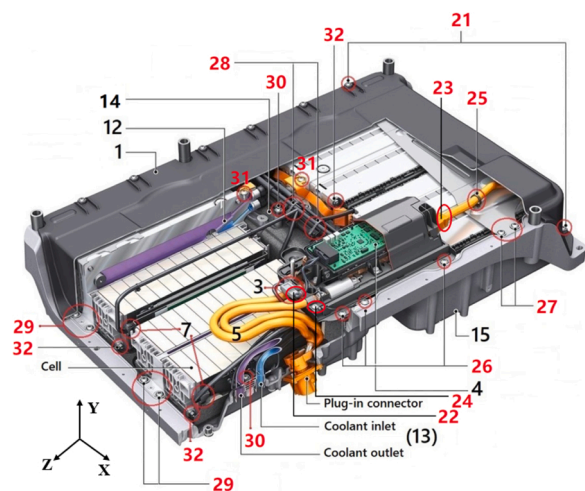


Fig. 7. The details of the HLBP. Modified from: [18].

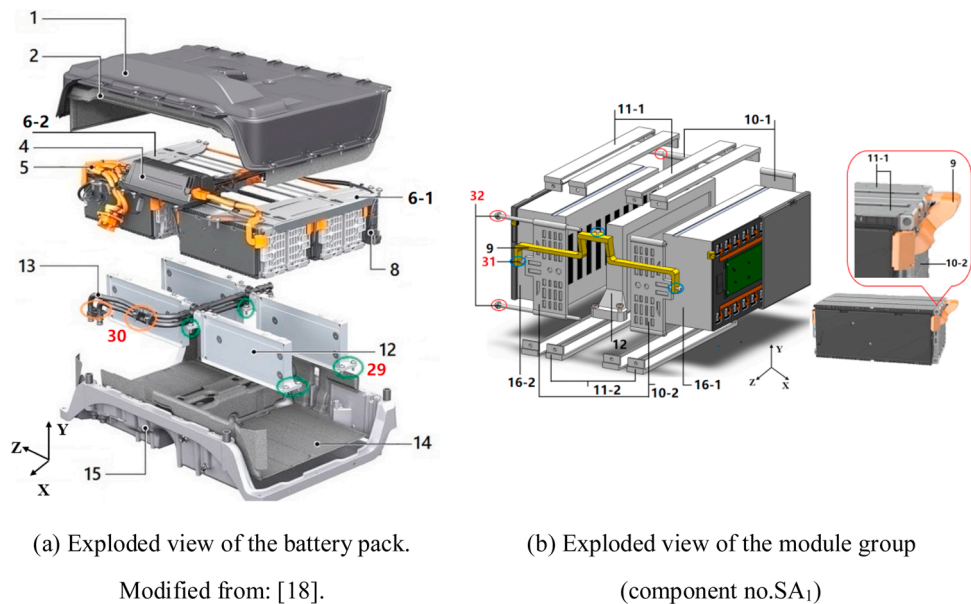


Fig. 8. Exploded view of the HLBP.

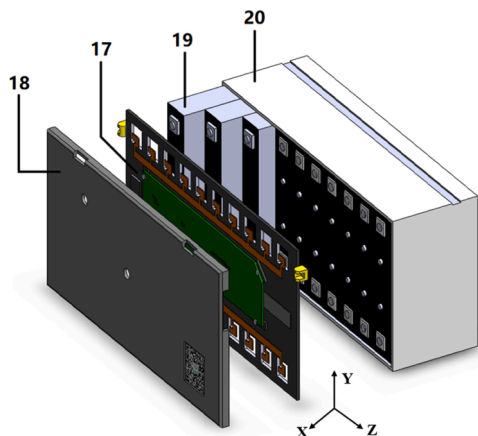


Fig. 9. Exploded view of the module (component no.16-1/16-2).

Table 7
Main components of the HLBP.

Component No.	Component name	Instance	quantity
1	Upper housing shell	upper_housing_shell	1
2	Upper insulator	upper_insulator	1
3	BJB connector	bjb_connector	1
4	Battery Junction Box (BJB)	battery_junction_box(BJB)	1
5	High Voltage cables and connectors	high_voltage_cables_and_connectors	1
6–1	Top transverse cover_right	top_transverse_cover_right	1
6–2	Top transverse cover_left	top_transverse_cover_left	1
7	Plug-in cable CMCs-BMC	plug-in_cable_CMCs-BMC	1
8	Battery Management Controller (BMC)	battery_management_controller(BMC)	1
SA ₁	Module group	module_group	4
9	Module connector	module_connector	4
10–1	Side module junction_right	side_module_junction_right	8
10–2	Side module junction_left	side_module_junction_left	8
11–1	Top pressing plate of module	top_pressing_plate_of_module	16
11–2	Bottom pressing plate of module	bottom_pressing_plate_of_module	16
12	Cooling plate	cooling_plate	4
13	Cooling pipe	cooling_pipe	2
14	Lower insulation	lower_insulator	1
15	Lower housing shell	lower_housing_shell	1
16–1	Module front	module_front	4
16–2	Module back	module_back	4
17	Cell Management Controller (CMC)	cell_management_controller(CMC)	8
18	Plastic cover	plastic_cover	8
SA ₂	Cell group	cell_group	8
R ₁	Restricted functional part pair	restricted_functional_part_pair	8
19	Cell	cell	96
20	Metal shell	metal_shell	8
21	Phillips screw of top housing shell	phillips_screw_of_top_housing_shell	18
22	Harness plug of BJB	harness_plug_of_BJB	1
23	Harness plug of CMCs	harness_plug_of_CMCs	1
24	Phillips screw of high Voltage cables	phillips_screw_of_high_voltage_cables	4
25	Cable tie of high voltage cable	cable_tie_of_high_voltage_cable	6
26	Phillips screw of junction box	phillips_screw_of_junction_box	6
27	Phillips screw of top	phillips_screw_of_top_transverse_cover	8

Table 7 (continued)

Component No.	Component name	Instance	quantity
28	transverse cover Cable tie of CMCs-BMC wire	cable_tie_of_CMCs-BMC_wire	6
29	Phillips screw of cooling plate	phillips_screw_of_cooling_plate	16
30	Phillips screw of cooling pipe	phillips_screw_of_cooling_pipe	7
31	Phillips screw of modules connector	phillips_screw_of_modules_connector	12
32	Bolt of side module junctions	blot_of_side_module_junctions	16

Table 8

The subassembly and restricted functional part pairs list of the HLBP.

No.	Components
SA ₁	9,10–1,10–2,11–1,11–2,12,16–1,16–2,31a,31b,31c, 32
SA ₂	1920
R ₁	17, SA ₂

Table 9

The connection list of the HLBP.

No.	Fastener	Functional Part	Connection
1	20		Welding
2	21		Phillips screw
3	22		Harness plug
4	23		Harness plug
5	24		Phillips screw
6	25		Cable tie
7	26		Phillips screw
8	27		Phillips screw
9	28		Cable tie
10	29		Phillips screw
11	30		Phillips screw
12	31		Phillips screw
13	32		Bolt
14		3,5	Bonding
15		R ₁ ,18	Cable buckle
16		17,SA ₂	Welding

5.2. DTP based on RBR

5.2.1. Partial destructive disassembly rules

Non-destructive disassembly rules for retired products are been researched in [34,39,51]. Based on the research conducted by Chen et al. [39], a set of partial destructive disassembly rules are formulated by using disassembly ontology and disassembly matrices, and described by SWRL language, as shown in Table 4. The functional parts can be replaced by subassemblies in the rules. Rules 1–23 are non-destructive disassembly rules and rules 24–26 are destructive disassembly rules. Rules 1–6 describe the disassembly sequence of fixing parts and functional parts with directly cover assembly relation. Rules 7–12 indicate the disassembly sequence of fixing parts and functional parts with fixed assembly relation. Rules 14–23 suggest describe the disassembly sequence of two functional parts with directly overlay assembly relation. Rules 24–26 describe the disassembly sequence of two functional parts with restricted assembly relation.

Table 10
Semantic assembly-relations of fixed parts in the battery pack.

Fastener no.	isDirectCoveredBy_plusX	isDirectCoveredBy_minusX	isDirectCoveredBy_plusY	isDirectCoveredBy_minusY	isDirectCoveredBy_plusZ	isDirectCoveredBy_minusZ
21				0		
22	2					
23						2
24				3		
25				2		
26				2		
27				5		
28				4		
29				7		
30				7		
31 _a	0					
31 _b		0				
31 _c		9				
32						

5.2.2. Iterative Generation method of feasible disassembly sequences

5.2.2.1. Notations. The symbols used in this section are listed, as shown in Table 5.

5.2.2.2. Definitions and assumptions. The terms and assumptions used in this section are defined as follows:

Definition 1. PRFP: A functional part that must be removed before the fixed part/functional part is disassembled.

Definition 2. PRF: A fixing part that must be removed before the functional part is disassembled.

Definition 3. Normal direction: The direction of the maximum plane perpendicular to the outer surface of the functional part. There is no normal direction for functional parts with irregular shapes, such as wiring harness, cable, and pipeline.

Definition 4. Retention principle: The smaller the change of disassembly directions, the lower the disassembly cost is. Therefore, when a functional part has multiple disassembly directions, only the disassembly direction in which the change of disassembly direction of this group of functional parts is minimal is retained.

Assumption 1. When a functional part encounters multiple fixed parts of priority disassembly, first remove the wiring harness, cable tie, card buckles, and then remove the threaded fasteners.

Assumption 2. The two functional parts connected by welding and bonding can be removed as a restricted functional part pair which is disassembled separately. In addition, subassemblies and restricted functional part pairs can be regarded as function parts for disassembly.

Assumption 3. It is assumed that the functional parts are disassembled in a direction other than -Y (gravity direction). Because the functional part below plays a supporting role on the part above, and the rotation of parts is ignored during disassembly.

Assumption 4. It is assumed that the normal direction of functional parts and the disassembly direction of fixed parts have been determined before disassembly.

6. Iterative Generation method of feasible disassembly sequence

The flow chart of generating feasible disassembly sequences of ATBs product is shown in Figs. 5 and 6. According to the iterative generation method of disassembly sequences and disassembly matrix, the disassembly direction of each part and the disassembly priority relation between all parts can be obtained by iteratively using the disassembly rules defined in Table 4. By synthesizing all disassembly directions and disassembly priority relations, multiple feasible disassembly sequences can be obtained. In Fig. 5, subassemblies that require disassembly of internal parts are represented by SA, and subassemblies that do not need to remove of internal parts are represented by FP. In Fig. 6, the face-to-face fit functional parts are disassembled preferentially along the normal direction to get closer to the actual disassembly.

6.1. Reasoning based on disassembly tool selection rules

The disassembly tools are selected by operations based on their experience knowledge. Tian et al. [52] used a hybrid calculation method to optimize the dismantlement efficiency of different disassembly tools for efficient selection of disassembly tools using bolts as the disassembly object. Jiang et al. [40] used first-order logic expression to describe part connection relations and the disassembly tool selection rules by taking electromechanical products as the disassembly object. In this paper, the disassembly tool selection rules approach from the literature [40] are used, and are described using the SWRL language. The syntax of SWRL

Table 11

Semantic assembly-relations of functional parts in the battery pack.

Functional -part no.}	normal Direction_X}	normal Direction _plusY}	normal Direction _Z}	isDirect CoveredBy _plusX}	isFixedBy _plusX}	isDirect Covered By_minusX}	isFixedBy_ minusX}	restrict _X}	isDirect Covered By_plusY}	isFixedBy_ plusY}	isDirect Covered By_minusY}	isFixedBy_ minusY}	restrict_Y}	isDirect Covered By_plusZ}	isFixedBy_ plusZ}	isDirect CoveredBy _minusZ}	isFixedBy_ _minusZ}	restrict_Z}
1	0	1	0	2		215			21					2		2		
2	0	1	0	1,4		1,4			1					1,6–1		1,6–2		
3	0	1	0		22				2							4		
4	0	1	0	2	22	2			2,5	26				3,5				23
5	0	0	0						3	24						4		
6–1	0	1	0						2	2527				SA ₁ ,7		2		
6–2	0	1	0						5	27				2		SA ₁ ,7		
7	0	1	0	SA ₁		SA ₁			4,6–1,6–2	28				SA ₁ ,6–2		SA ₁ ,6–1		
8	1	0	0	SA ₁		15			6–1	–				14		15		
SA ₁	0	1	0	7,13,14		7,8,14			4,6–1,6–2	29				6–2,7,13,14		6–1,7,13,14		
9	0	0	0		31a		31b			31c						10–2,11–1		
10–1	0	0	1	16–1,11–1, 11–2		16–2,11–1, 11–2	32		11–1					16–1,16–2		11–1, 11–2,12		
10–2	0	0	1	16–1,11–1, 11–2		16–2,11–1, 11–2	32		11–1					9,11–1, 11–2,12		16–1,16–2		
11–1	0	1	0	16–1,16–2		16–1,16–2	32							10–1		10–2		
11–2	0	1	0	16–1,16–2		16–1,16–2	32		10–1,10–2, 16–1,16–2					10–1		10–2		
12	1	0	0	16–1,10–1		16–2,10–2	32							10–1		10–2		
13	0	0	0	14		SA ₁			4,7	30				SA ₁		SA ₁		
14	0	1	0	SA ₁ ,15		SA ₁ ,1315			7,8, SA ₁ ,13					SA ₁ ,15		8,SA ₁ ,15		
15	0	1	0	1,8,14		14			1,214					814		14		
16–1	1	0	0	11–1,11–2	31a	12,11–1,11–2			11–1					10–2,11–1,11–2		10–1,11–1,11–2		
16–2	1	0	0	12,11–1,11–2		11–1,11–2	31b		11–1					10–2,11–1,11–2		10–1,11–1,11–2		

Table 12
Semantic assembly-relations of fixed parts in the battery module.

Fastener no.	isDirectCoveredBy_plusX	isDirectCoveredBy_minusX	isDirectCoveredBy_plusY	isDirectCoveredBy_minusY	isDirectCoveredBy_plusZ	isDirectCoveredBy_minusZ
20					17	17

rules is as follows: $A_1, A_2, \dots, A_m \rightarrow B_1, B_2, \dots, B_n$. Where $A_1, A_2, \dots, A_m$ are the conditions of the rule, $B_1, B_2, \dots, B_n$ are the results of the rule. The SWRL expression shown by formula (3) is used to express disassembly knowledge for selecting disassembly tools. The disassembly tools and connection modes of ATBs are shown in Table 6.

$$\begin{aligned} & \text{Class(Fastener/FunctionalPart, } p_i), \text{ Class(Fastener/FunctionalPart, } p_j), \text{ Class} \\ & (\text{Connection, } c_{ij}), \text{ has_connection}(p_i, c_{ij}), \text{ has_connection}(p_j, c_{ij}) \\ & \rightarrow \text{is_removed_with}(c_{ij}, t_k) \end{aligned} \quad (3)$$

In the formula (3), the parts p_i and p_j belong to the part class Fastener/FunctionalPart, and the connection relation c_{ij} belongs to the connection relation class Connection. If p_i has a connection relation c_{ij} and p_j also has a connection relation c_{ij} , then it is inferred that the disassembly tool t_k is used to remove the connection relation c_{ij} . The instance of the rule is shown in formula (4). The rule for the disassembly tool from knowledge is that if the connection relation between the two functional parts is bonding, the 'Slotted screwdriver' should be used to dismantle.

$$\begin{aligned} & \text{Class(FunctionalPart, } p_i), \text{ Class(FunctionalPart, } p_j), \text{ Class(Bonding, } c_{ij}), \text{ has_} \\ & \text{connection}(p_i, c_{ij}), \\ & \text{has_connection}(p_j, c_{ij}) \rightarrow \text{is_removed_with}(c_{ij}, \text{Slotted_screw_driver}) \end{aligned} \quad (4)$$

7. Case study and discussion

7.1. Case study

In this paper, a hybrid lithium-ion battery pack (HLBP) is used to test the presented method [53]. It is assumed that the battery pack, modules, and cells have QR codes attached to them. Case retrieval and matching are carried out after identifying the QR codes information. Assuming that there is no matching case in the case base, the DTP based on RBR is performed.

The HLBP consists of 8 modules, battery junction box (BJB), coolant system and battery management controller (BMC). Each module contains 12 cells, a total of 96 cells. Figs. 7 and 8 provide details of the HLBP being disassembled into modules. Fig. 9 is the explosion diagram of the module disassembled into cells, and its components are identified. Table 7 is the bill of materials, which lists the name and quantity of each part of HLBP. Moreover, the HLBP has a subassembly SA_1 . The module has a subassembly SA_2 and a restricted functional part pair R_1 . Table 8 lists the components of each subassembly and restricted functional part pair. The connections between components are shown in Table 9. After describing the components of the HLBP, the semantic assembly relations of the battery pack/module are given as shown in Tables 10,11,12 and 13.

To demonstrate the approach presented in this paper, the case uses Protégé 5.2.0 as a modeling tool to build domain ontology and ontology rules, as shown in Fig. 10 and Fig. 11. Area (1) in Fig. 11 lists disassembly tool selection rules and area (2) lists some partial destructive rules. The disassembly task planning process of the HLBP is shown in Fig. 12, which is similar to that of the module. Table 14 and 15 show the planning results of battery pack/module, and the results are practical. Among them, Table 14 lists only three feasible candidate disassembly schemes.

7.2. Discussion

The common destructive connection modes in ATBs include cable tie connection, interference fit, welding and bonding. The two parts connected by cable tie are classified as fixing assembly relation, and the cable tie is first removed as a fixing part. The pressure part with interference fit assembly relation with the functional part is also

Table 13
Semantic assembly-relations of functional parts in the battery module.

Functional partno.	normal Direction_X	normal Direction_plusY	normal Direction_Z	isDirect Covered By_plusX	isFixedBy_plusX	isDirect CoveredBy_minusX	isFixedBy_minusX	restrict_X	isDirect Covered By_plusY	isFixedBy_plusY	isDirect Covered By_minusY	isFixedBy_minusY	restrict_Y	isDirect Covered By_plusZ	isFixedBy_plusZ	isDirect Covered By_minusZ	isFixedBy_minusZ	restrict_Z
17	1	0	0			SA ₂												
18	1	0	0			SA ₂												
19	0	0	1		20	R ₁	20							20			20	
R ₁	1	0	0	18														
SA ₂	1	0	0	17				17										

disassembled as a fixing part in advance. The restricted functional part pair connected by welding or bonding are categorized as restricted assembly relation and there is no sequence difference between them when conducting the disassembling planning. Other non-destructive connections are referred to Reference [39]. The disassembly matrices in this paper determine the disassembly priority of parts according to the assembly relations. It only needs to describe the parts number with assembly relations in six disassembly directions for each part, without the need to describe the disassembly priority relations and connection relations between all parts. This method greatly reduces the computation space and time. In each disassembly step listed in Tables 14 and 15, one or more components are removed within one step. This is because these component do not interfere with each other, thus can be disassembled in parallel.

In addition, taking the battery in this paper as an example, the iterative generation method proposed in this paper is compared with the method from reference [39]. The disassembly sequences generated for the battery pack are consistent. The disassembly sequences generated by the battery module have only different directions, as shown in Table 16. It can be seen that the rule of priority disassembly along the normal direction of the parts not only conforms to ergonomics requirement, but also effectively eliminates some unrealistic disassembly sequences, instead of generating all possible disassembly sequences. Therefore, this iterative generation method of disassembly sequences essentially generates some higher quality executable parallel disassembly sequences.

To facilitate the intelligent disassembly of ATBs, the detachability of components should be included in the assessment target at the design stage. For example, considering modular design and avoiding the use of welding, bonding and other connections that are not easily removed by robots [54]. At the battery design stage, the disassembly scheme may be planned in advanced and stored in a label (such as RFID) of the battery [10]. The information stored in the label can be re-edited and modification traces are recorded. This method not only immensely reduces the time and cost of generating a battery disassembly scheme at the end-of-life stage, but also speeds up the process of intelligent disassembly of battery.

8. Conclusion and future work

Many kinds of UTD parts in ATBs need to be disassembled in a partial destructive way. In order to reuse the previous disassembly knowledge and promote the intelligent disassembly of ATBs, a method of DTP based on ontology and partial destructive rules is proposed. In this method, ontology and matrices are used for knowledge modelling, and the disassembly tools and disassembly priority relations between parts are represented by ontology rules. CBR/RBR method is used as the reuse and reasoning mechanism of disassembly knowledge to generate DTP of ATBs. The CBR method is used to obtain the best disassembly solution, while the RBR method is used to generate some higher quality disassembly solutions. The contribution of this work is that the proposed method can reuse the previous disassembly knowledge to rapidly generate good quality disassembly schemes. Moreover, it is easy to integrate into the Internet of Things platform in future with the employment of OWL (Web Ontology Language) modeling, which provides technical support for the intelligent disassembly of ATBs.

In the future work, there are several aspects to be improved:

- The disassembly matrices in this paper have to be input manually, which affects the planning efficiency, and an automatic input mode can be studied later. In addition, ontology and disassembly rules need to be extended to render them useful, and more existing disassembly knowledge can be added to the case library and rule library to improve the accuracy of disassembly planning. In the future, machine learning methods can be used to learn from disassembly operations and gradually abandon manual rules to improve disassembly efficiency [55].

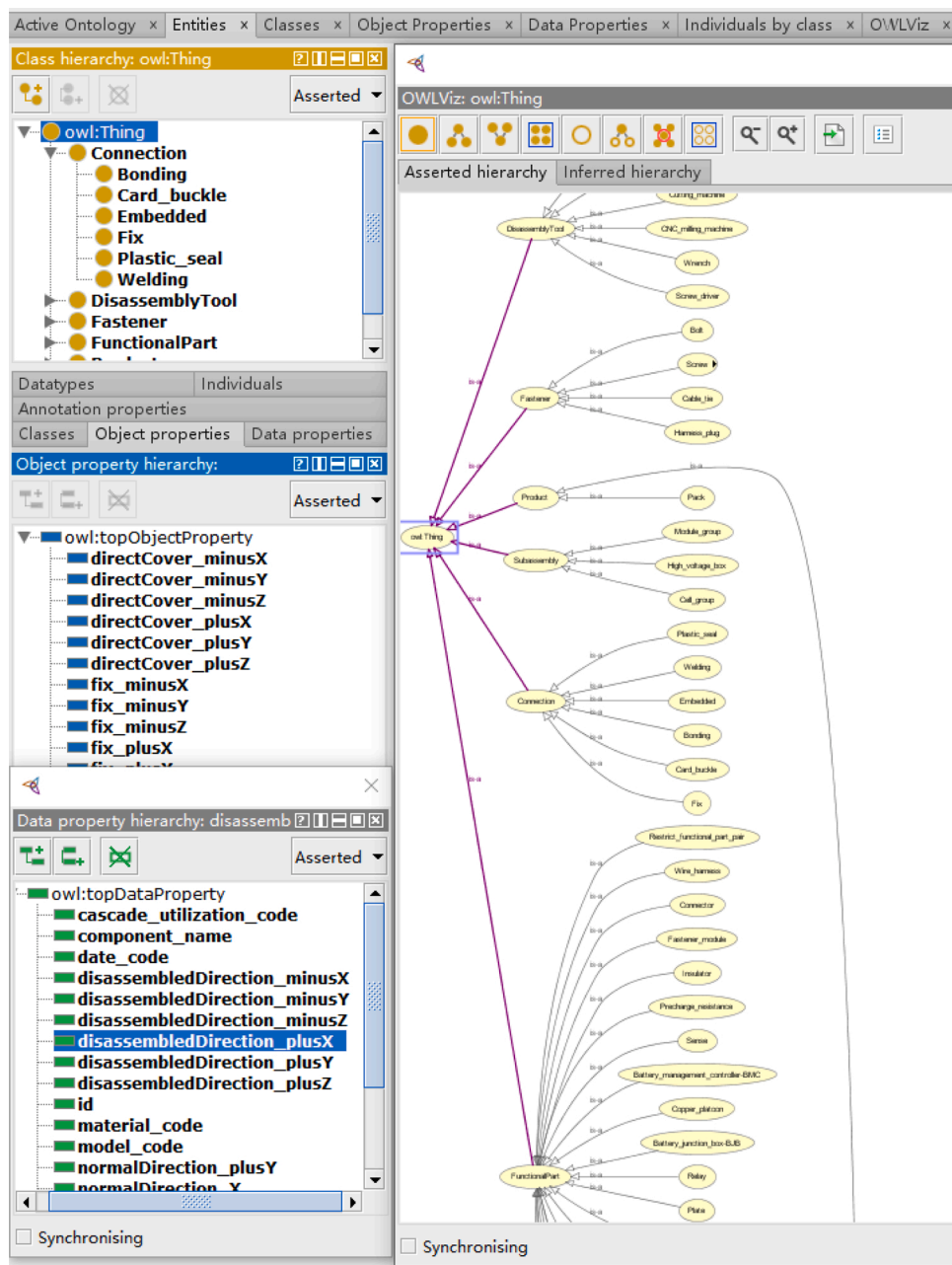


Fig. 10. Screenshot of the disassembly ontology in Protégé.

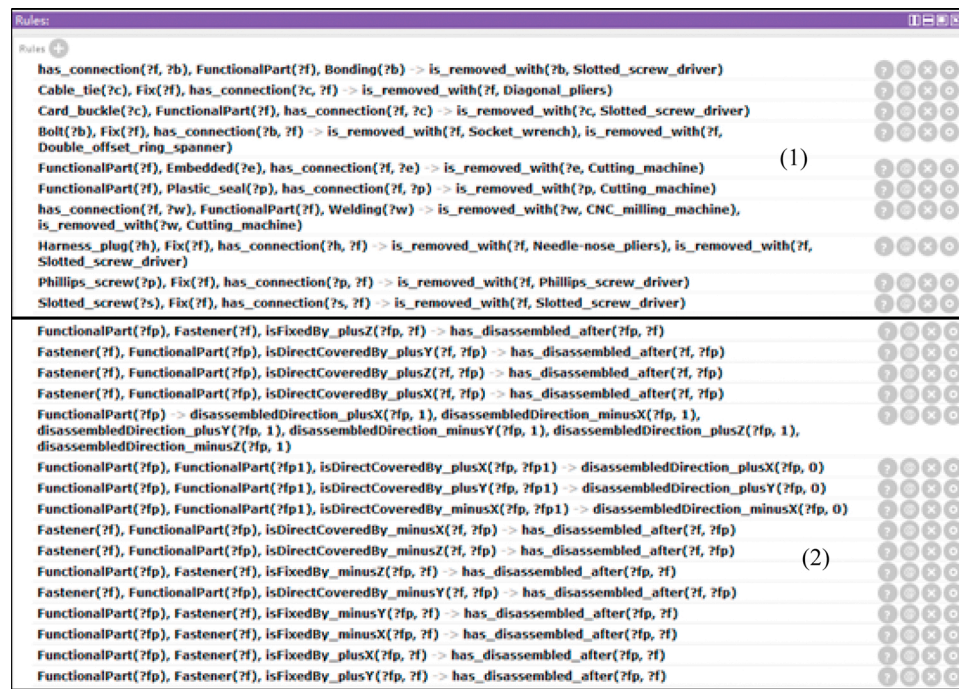


Fig. 11. A part of ontology rules.

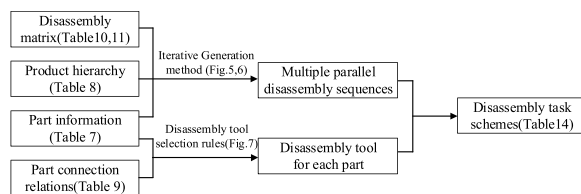


Fig. 12. The process diagram of DTP based on RBR for the battery pack.

- The RBR method may generate multiple feasible parallel disassembly schemes. Subsequent work may use an intelligent algorithm to obtain the optimal or near-optimal disassembly scheme from economic, technical and environmental perspectives.
- In the future, a disassembly planning system of ATBs can be built with user friendly interface to facilitate users to access decision support services to guide disassembly. The shared disassembly knowledge can be used in disassembly design, automatic disassembly and human-machine cooperative disassembly.

Data availability

No data was used for the research described in the article.
Data will be made available on request.
All data is within the manuscript and figure
Data will be made available on request.
No data was used for the research described in the article.
The authors do not have permission to share data.

Table 14

Several feasible disassembly schemes for the battery pack.

No.	Disassembly sequence	Disassembly tool number
1	21(+Y),1(+Y),2(+Y),{22(+X),25(+Y)}, {23(-Z),26(+Y)},3(+Y),24(+Y), 5(+Y),27(+Y),{4(+Y),6-1(+Y),6-2(+Y)},28(+Y),{7(+Y),8(+Y)}, {29(+Y),30(+Y)}, {SA ₁ (+Y),13(+Y)},14(+Y),15(+Y),{31a(+X),31b(-X),31c(+Y)}, 9(+Y),32(-X),{11-1(+Y),12(+Y)}, {10-1(+Y),10-2(+Y),16-1(+Y),16-2(+Y)},11-2(+Y)	T4,-,-,{T5/T3,T6}, {T5/T3,T4},T3,T4,-,T4,{-,,-},T6, {-,,-},{T4,T4},{-,,-},-,{T4,T4,T4},-, T1/T2,{-,,-},{-,,-},-
2	21(+Y),1(+Y),2(+Y),{22(+X),25(+Y)}, 3(+Y),{23(-Z),24(+Y),26(+Y)}, 5(+Y), {4(+Y),27(+Y)}, {6-1(+Y),6-2(+Y),28(+Y)}, {7(+Y),8(+Y)}, {29(+Y),30(+Y)}, {SA ₁ (+Y),13(+Y)},14(+Y),15(+Y),{31a(+X),31b(-X),31c(+Y)}, 9(+Y),32(-X),{11-1(+Y),12(+Y)}, {10-1(+Y),10-2(+Y)}, {16-1(+Y),16-2(+Y)},11-2(+Y)	T4,-,-,{T5/T3,T6},T3,{T5/T3,T4,T4},-,{-,T4},{-,T6},{-,,-}, {T4,T4},{-,,-},-,{T4,T4,T4},-,T1/T2, {-,,-},{-,,-},{-,,-},-
3	21(+Y),1(+Y),2(+Y),22(+X), 3(+Y),{25(+Y),26(+Y)}, {24(+Y),23(-Z)}, 5(+Y), {4(+Y),27(+Y)}, {6-1(+Y),6-2(+Y)}, {8(+Y)28(+Y)},7(+Y),{29(+Y),30(+Y)}, {SA ₁ (+Y),13(+Y)},14(+Y),15(+Y),{31a(+X),31b(-X),31c(+Y)}, 9(+Y),32(-X),{11-1(+Y),12(+Y)}, {10-1(+Y),10-2(+Y)}, {16-1(+Y),16-2(+Y)},11-2(+Y)	T4,-,-,T5/T3,T3,{T6,T4},{T4,T5/T3},-,{-,T4},{-,,-},{-,T6},-,{-,T4,T4}, {-,,-},-,{T4,T4,T4},-,T1/T2,{-,,-}, {-,,-},{-,,-},-

Note: the symbol "-" in the table indicates that the part does not need to be disassembled with tools.

Table 15

The list of disassembly task for the battery module.

Disassembly sequence	Disassembly tool number
{18(+X),R ₁ (-X)},{17(+X),SA ₂ (-X)},{20(+Z,-Z),19(+Z)}	T3,T7/T8,T8,-

Table 16

The disassembly sequences obtained by two iterative generation methods for the battery module.

Item.	Disassembly sequence	Disassembly tool number
The proposed method	{18(+X),R ₁ (-X)},{17(+X),SA ₂ (-X)},{20(+Z,-Z),19(+Z)}	T3,T7/T8,T8,-
The method of [39]	{18(+Y),R ₁ (+Y)},{17(+Y),SA ₂ (+Y)},{20(+Z,-Z),19(+Y)}	

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Harper G, Sommerville R, Kendrick E, Driscoll L, Slater P, Stolkin R, et al. Recycling lithium-ion batteries from electric vehicles. *Nature* 2019;575:75–86. <https://doi.org/10.1038/s41586-019-1682-5>.
- Huaxin Securities. In-depth report of New Energy vehicle Industry in 2021, <http://new.qq.com/omn/20210812/20210812A01US200.html>; 2021 [accessed 20 August 2021].
- Chen M, Ma X, Chen B, Arsenault R, Karlson P, Simon N, et al. Recycling end-of-Life electric vehicle lithium-ion batteries. *Joule* 2019;3:2622–46. <https://doi.org/10.1016/j.joule.2019.09.014>.
- Hellmuth JF, DiFilippo NM, Jouaneh MK. Assessment of the automation potential of electric vehicle battery disassembly. *Int J Ind Manuf Syst Eng* 2021;59:398–412. <https://doi.org/10.1016/j.jmsy.2021.03.009>.
- Kootstra MA, Tong S, Park JW. Photovoltaic grid stabilization system using second life lithium battery: photovoltaic grid stabilization. *Int J Energy Res* 2015;39. <https://doi.org/10.1002/er.3310>.
- Hakim I, Costa PD. Is electric vehicles battery recovery a source of cost or profit? *Automob. Revolut.*, Geneva, Switzerland. Cham: Springer; 2017. p. 117–34.
- Zhang H, Huang J, Hu R, Zhou D, Khan H ur R, Ma C. Echelon utilization of waste power batteries in new energy vehicles: review of Chinese policies. *Energy* 2020;206:118178. <https://doi.org/10.1016/j.energy.2020.118178>.
- Tian G, Liu Y, Ke H, Chu J. Energy evaluation method and its optimization models for process planning with stochastic characteristics: a case study in disassembly decision-making. *Comput Ind Eng* 2012;63:553–63. <https://doi.org/10.1016/j.cie.2011.08.011>.
- Liu J, Zhou Z, Pham DT, Xu W, Cui J, Yang C. Service platform for robotic disassembly planning in remanufacturing. *Int J Ind Manuf Syst Eng* 2020;57:338–56. <https://doi.org/10.1016/j.jmsy.2020.10.005>.
- Weyrich M, Wang Y. Architecture design of a vision-based intelligent system for automated disassembly of E-waste with a case study of traction batteries. *IEEE Conference on Emerging Technologies & Factory Automation. IEEE* 2013. <https://doi.org/10.1109/ETFA.2013.6648043>.
- Zhou L, Garg A, Zheng J, Gao L, Oh K-Y. Battery pack recycling challenges for the year 2030: recommended solutions based on intelligent robotics for safe and efficient disassembly, residual energy detection, and secondary utilization. *Energy Storage Mater* 2020;3:e190. <https://doi.org/10.1002/est2.190>.
- Song X, Zhou W, Pan X, Feng K. Disassembly sequence planning for electro-mechanical products under a partial destructive mode. *Assem Autom* 2014. <https://doi.org/10.1108/AA-01-2013-006>. 34.
- Tang Y, Zhou M, Zussman E, Caudill R. Disassembly modeling, planning, and application. *Int J Ind Manuf Syst Eng* 2002;21(3):200–17. [https://doi.org/10.1016/S0278-6125\(02\)80162-5](https://doi.org/10.1016/S0278-6125(02)80162-5).
- Lambert AJD. Disassembly sequencing: a survey. *Int J Prod Res* 2003;41:3721–59. <https://doi.org/10.1080/0020754031000120078>.
- Ong SK, Chang M, Nee A. Product disassembly sequence planning: state-of-the-art, challenges, opportunities and future directions. *Int J Prod Res* 2021;1–16. <https://doi.org/10.1080/00207543.2020.1868598>.
- Lai X, Huang Y, Deng C, Gu H, Han X, Zheng Y, et al. Sorting, regrouping, and echelon utilization of the large-scale retired lithium batteries: a critical review. *Renewable Sustainable Energy Rev* 2021;146:111162. <https://doi.org/10.1016/j.rser.2021.111162>.
- Herrmann C, Raatz A, Mennenga M, Schmitt J, Andrew S. Assessment of automation potentials for the disassembly of automotive Lithium ion battery systems. Berkeley, USA: Berlin, Heidelberg: Springer; 2012. p. 149–54. <https://doi.org/10.1007/978-3-642-29069-5>.
- Alfaro-Algaba M, Ramirez FJ. Techno-economic and environmental disassembly planning of lithium-ion electric vehicle battery packs for remanufacturing. *Resour Conserv Recycl* 2020;154:104461. <https://doi.org/10.1016/j.resconrec.2019.104461>.
- Yang J, Gu F, Guo J. Environmental feasibility of secondary use of electric vehicle lithium-ion batteries in communication base stations. *Resour Conserv Recycl* 2020;156:104713. <https://doi.org/10.1016/j.resconrec.2020.104713>.
- Herrmann C, Raatz A, Andrew S, Schmitt J. Scenario-based development of disassembly systems for automotive Lithium ion battery systems. *Adv Mater Res* 2014;907:391–401. <https://doi.org/10.4028/www.scientific.net/AMR.907.391>.
- Schwarz TE, Rübenbauer W, Rutrecht B, Pomberger R. Forecasting real disassembly time of industrial batteries based on virtual MTM-UAS data. *Procedia Cirp* 2018;69:927–31. <https://doi.org/10.1016/j.procir.2017.11.094>.
- Schäfer J, Singer R, Hofmann J, Fleischer J. Challenges and solutions of automated disassembly and condition-based remanufacturing of lithium-ion battery modules for a circular economy. *Procedia Manuf* 2020;43:614–9. <https://doi.org/10.1016/j.promfg.2020.02.145>.
- Gerbers R, Wegener K, Dietrich F, Dröder K. Safe, flexible and productive human-robot-collaboration for disassembly of lithium-ion batteries. In: Kwade A, Diekmann J, editors. *Recycl. Lithium-ion batter. LithoRec Way*. Cham: Springer International Publishing; 2018. p. 99–126.
- Tan W, Chin C, Garg A, Gao L. A hybrid disassembly framework for disassembly of electric vehicle batteries. *Int J Energy Res* 2021;45. <https://doi.org/10.1002/er.6364>.
- Choux M, Bigorra E, Tyapin I. Task planner for robotic disassembly of electric vehicle battery pack. *Metals* 2021;11:387–404. <https://doi.org/10.3390/met11030387>.
- Yuan L, Cui J, Zhang X, Liu J. Framework and enabling technologies of cloud robotic disassembly. *Procedia Comput Sci* 2020;176:3673–81. <https://doi.org/10.1016/j.procs.2020.09.019>.
- Wegener K, Andrew S, Raatz A, Dröder K, Herrmann C. Disassembly of electric vehicle batteries using the example of the audi Q5 hybrid system. *Procedia Cirp* 2014;23:155–60. <https://doi.org/10.1016/j.procir.2014.10.098>.
- Ke Q, Zhang P, Zhang L, Song S. Electric vehicle battery disassembly sequence planning based on frame-subgroup structure combined with genetic algorithm. *Front Mech Eng China* 2020;6:576642. <https://doi.org/10.3389/fmech.2020.576642>.
- Berg LP, Behdad S, Vance JM, Thurston D. Disassembly sequence evaluation: a user study leveraging immersive computing technologies. *J Comput Inf Sci Eng* 2015;15. <https://doi.org/10.1115/1.4028857>.
- Zhu B, Sarigecili MI, Roy U. Disassembly information model incorporating dynamic capabilities for disassembly sequence generation. *Robot Comput-Integr Manuf* 2013;29:396–409. <https://doi.org/10.1016/j.rcim.2013.03.003>.
- Zhou Z, Liu J, Pham DT, Xu W, Ramirez FJ, Ji C, et al. Disassembly sequence planning: recent developments and future trends. *Proc Inst Mech Eng Part B J Eng Manuf* 2018;233:1450–71. <https://doi.org/10.1177/0954405418789975>.
- Ren Y, Zhang C, Zhao F, Xiao H, Tian G. An asynchronous parallel disassembly planning based on genetic algorithm. *Eur J Oper Res* 2018;269:647–60. <https://doi.org/10.1016/j.ejor.2018.01.055>.
- Tian G, Ren Y, Feng Y, Zhou M, Zhang H, Tan J. Modeling and planning for dual-objective selective disassembly using AND/OR graph and discrete artificial bee colony. *Ieee Trans Ind Inform* 2018;15:2456–68. <https://doi.org/10.1109/TII.2018.2884845>.
- Smith SS, Chen W-H. Rule-based recursive selective disassembly sequence planning for green design. *Int J Adv Sci Eng Inf Technol* 2011;25:77–87. <https://doi.org/10.1016/j.aei.2010.03.002>.
- Rickli JL, Camello JA. Multi-objective partial disassembly optimization based on sequence feasibility. *Int J Ind Manuf Syst Eng* 2013;32(1):281–93. <https://doi.org/10.1016/j.jmsy.2012.11.005>.
- Tao F, Bi L, Zuo Y, Nee AYC. Partial/Parallel disassembly sequence planning for complex products. *J Manuf Sci Eng-Trans Asme* 2018;140:011016. <https://doi.org/10.1115/1.4037608>.
- Zhu B, Roy U. Ontology-based disassembly information system for enhancing disassembly planning and design. *Int J Adv Manuf Technol* 2015;78:1595–608. <https://doi.org/10.1007/s00170-014-6704-8>.
- Zhu B, Roy U. Modeling and validation of a web ontology language based disassembly planning information model. *J Comput Inf Sci Eng* 2018;18:021015. <https://doi.org/10.1115/1.4039849>.

- [39] Chen S, Yi J, Jiang H, Zhu X. Ontology and CBR based automated decision-making method for the disassembly of mechanical products. *Int J Adv Sci Eng Inf Technol* 2016;30:564–84. <https://doi.org/10.1016/j.aei.2016.06.005>.
- [40] Jiang H, Yi J, Zhu X, Li Z. Generating disassembly tasks for selective disassembly using ontology-based disassembly knowledge representation. *Assem Autom* 2017;38:113–24. <https://doi.org/10.1108/aa-04-2016-034>.
- [41] Yang L, Cormican K, Yu M. Ontology-based systems engineering: a state-of-the-art review. *Comput Ind* 2019;111:148–71. <https://doi.org/10.1016/j.compind.2019.05.003>.
- [42] Zhong B, Wu H, Li H, Sepasgozar S, Luo H, He L. A scientometric analysis and critical review of construction related ontology research. *Autom Constr* 2019;101:17–31. <https://doi.org/10.1016/j.autcon.2018.12.013>.
- [43] He Y, Hao C, Wang Y, Li Y, Wang Y, Huang L, et al. An ontology-based method of knowledge modelling for remanufacturing process planning. *J Clean Prod* 2020;258:120952. <https://doi.org/10.1016/j.jclepro.2020.120952>.
- [44] Park G, Benedictos J, Lee C, Wang M. Ontology-based Fuzzy-CBR support system for ship's collision avoidance, Hong Kong. 2007. <https://doi.org/10.1109/ICMLC.2007.4370448>.
- [45] Bouhana A, Zidi A, Fekih A, Chabchoub H, Abed M. An ontology-based CBR approach for personalized itinerary search systems for sustainable urban freight transport. *Expert Syst Appl* 2015;42:3724–41. <https://doi.org/10.1016/j.eswa.2014.12.012>.
- [46] Camarillo A, Ríos J, Althoff K-D. Knowledge-based multi-agent system for manufacturing problem solving process in production plants. *Int J Ind Manuf Syst Eng* 2018;47:115–27. <https://doi.org/10.1016/j.jmsy.2018.04.002>.
- [47] Yang S-Y, Hsu C-L. An ontological Proxy Agent with prediction, CBR, and RBR techniques for fast query processing. *Expert Syst Appl* 2009;36:9358–70. <https://doi.org/10.1016/j.eswa.2009.01.011>.
- [48] Xu F, Liu X, Chen W, Zhou C, Cao B. Ontology-based method for fault diagnosis of loaders. *Sensors* 2018;18. <https://doi.org/10.3390/s18030729>.
- [49] Electric Vehicles Research. All EV batteries born after August 2018 in China will have unique IDs, <https://www.electricvehiclesresearch.com/articles/15455/all-ev-batteries-born-after-august-2018-in-china-will-have-unique-ids/>;2018 [accessed 4 November 2020].
- [50] Chen S, Yi J, Zhu X, Jiang H, Ju W. RFID-based integrated method for electromechanical products disassembly decision-making. *Int J Comput Integr Manuf* 2016;30:229–54. <https://doi.org/10.1080/0951192x.2016.1145807>.
- [51] Torres F, Puente ST, Aracil R. Disassembly planning based on precedence relations among assemblies. *Int J Adv Manuf Technol* 2003;21:317–27. <https://doi.org/10.1007/s001700300037>.
- [52] Tian G, Qiang T, Chu J, Xu G, Zhou W. Efficiency optimization for disassembly tools via using NN-GA approach. *Math Probl Eng* 2013;2013:1–8. <https://doi.org/10.1155/2013/173736>.
- [53] Audi AG. The video of Audi-A3-Sportback-e-tron, https://www.audi-technology-portal.de/en/mobility-for-the-future/audi-future-lab-tron-experience_en/audi-a3-e-tron_en/; 2020 [accessed 04 November 2020].
- [54] Chitre A, Freake D, Lander L, Edge J, Titirici M. Towards a more sustainable lithium-ion battery future: recycling LIBs from electric vehicles. *Batter Supercaps* 2020. <https://doi.org/10.1002/batt.202000146>.
- [55] Chen X., Jia S., Xiang Y. A review: Knowledge reasoning over knowledge graph. *Expert Syst Appl* ;141:112948. <https://doi.org/10.1016/j.eswa.2019.112948>.